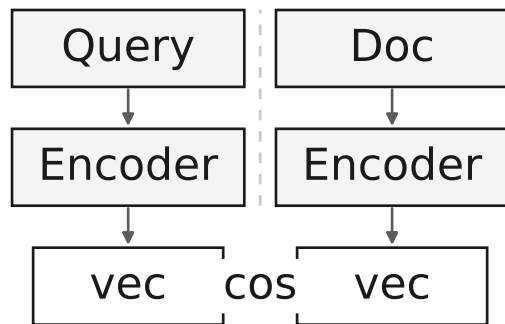


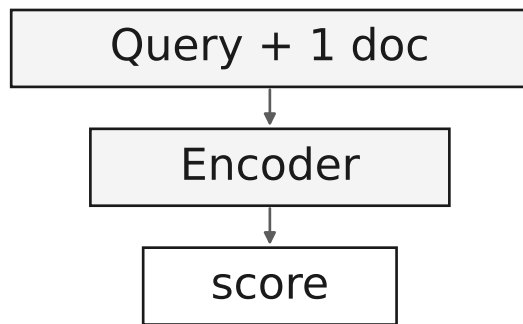
Bi-encoder

embedding pipeline



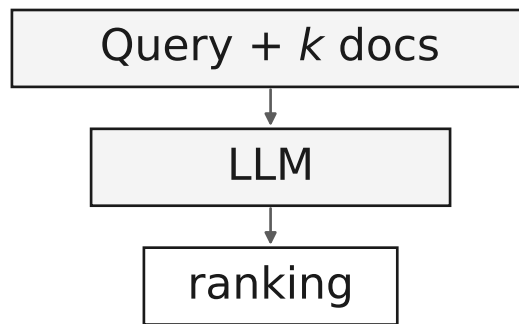
Cross-encoder

pairwise reranker



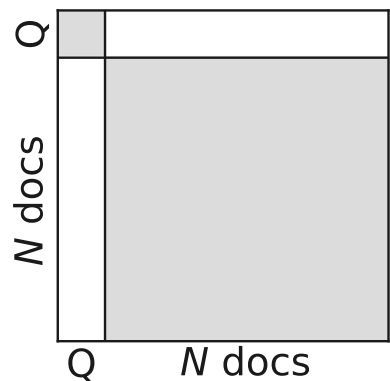
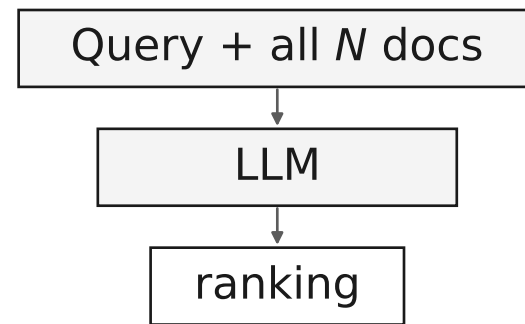
LLM listwise

reranker over top- k



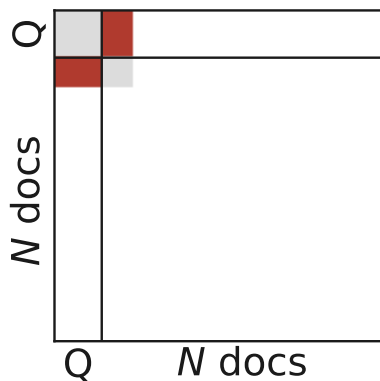
LLM corpus-in-context

generative retrieval



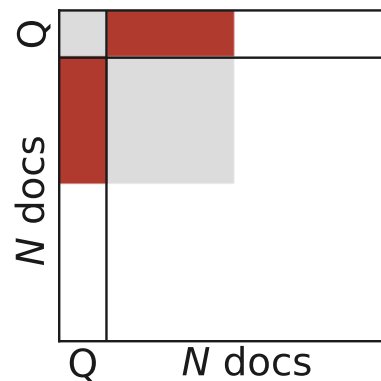
never jointly

N vector comparisons
per query



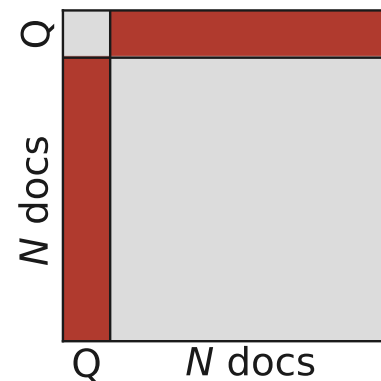
one document at a time

k pairwise passes
per query



k documents at once

one pass over k docs
per query



the whole corpus at once

one pass over N docs
per query



self-attention



query-document



no attention